



Lab 09 + 10 - Mini projects

Name your solution **Week0910**

A. Project StartupSponsors

Scenario

The company **AlphaTech** opens its monthly internal **Hackathon Challenge**. The employees of the company will form teams to participate. Each team will submit its project for getting sponsors from the company.

There are two types of project:

- **Idea:** An initial idea is all that the team has. This project needs time to do research and development. It will propose the months that are needed.
 - For example: Team A has an ideas of developing a website for renting old computers. This project needs **6** months of developing.
 - The company can support each month **2.200\$** for development.
 - As a result, the company needs to grant a sponsor of $6 * 2.200 = 13.200\$$.

- **MVP:** This project has some basic development and reaches MVP state (minimal viable product). It will need money for running in production and its marketing campaigns. The team will propose the months for running the product in production together with the months they want to execute their marketing campaigns.
 - For example: Team B has developed an MVP version of the project **Flower Fast Delivery**. It wants to runs the website for **8** months and use **4** months to do some marketing campaigns.
 - The company can support each month **930\$** for running the product.
 - To support the website, the company will hire a marketing team in the market with the cost of **500\$** each month.
 - As a result, the company needs to grant a sponsor of $8 * 930 + 4 * 500 =$ **9.440\$**

Requirements

- Read all project proposals from the input file: **MayProposals.txt**

Idea: SuperTech, DevMonths=6 Mvp: FutureCar, RunMonths=6, Marketing=3 Idea: VirtualHouse, DevMonths=7 Idea: NightVision, DevMonths=4 Mvp: SmartKey, RunMonths=4, Marketing=5

- Print out in table format all the projects' sponsor information, sorted using the amount of money sponsored in ascending order
- Print out the total sponsor amount the company must give away this month.

Sample output

💡 Sample data is just for presentation illustration only, **not actual data!**

```
Kế hoạch tài trợ dự án startup nội bộ công ty AlphaTech Tháng: 05/2024 STT |
Loại dự án | Tên dự án | Thông tin | Kinh phí -----
----- 1 | Ý
tưởng | NightVision | Phát triển: 6 tháng | $120,000 2 | Sản phẩm | SmartKey
| Vận hành: 6 tháng. Marketing: 3 tháng | $150,000 3 | Ý tưởng | VirtualHouse
| Phát triển: 7 tháng | $160,000 4 | Ý tưởng | SuperTech | Phát triển: 4
tháng | $280,000 5 | Sản phẩm | FutureCar | Vận hành: 4 tháng. Marketing: 5
tháng | $335,000 Tổng kinh phí tài trợ: $800,000
```

B. Project TaxCollecting

Scenario

In Ho Chi Minh city, the Ben Thanh market has a lot of booths for rent. The renters usually use the stall for selling food, clothes and jewelry.

- For every square meter of the stall's area, the renters must pay an additional amount of **560.000** dong.
- The food booth must pay additional money for using cold storage service for food preservation. The amount is different for each booth based on its usage.
- The jewelry booth must pay additional money for security service (different for each booth).

Requirements

- Read all the information of the rented booths from the file **RentMay24.txt**.

```
Food ID=A724, Area=78, ColdStorage=3000000 Clothes ID=B419, Area=33 Jewelry
ID=C682, Area=55, SecurityFee=7000000 Clothes ID=B331, Area=38 Food ID=A903,
Area=60, ColdStorage=5000000
```

- Print out all renting info sorted by the rent price in ascending order. This report will help the owner to collect the renting payment in this month.
- Print out the total amount of renting the owner should receive after collecting all the renting payment.

Sample output

 Sample data is just for presentation illustration only, **not actual data!**

```
Bao cao thu tien thue mat bang cho Ben Thanh Thang: 05/2024 STT | Ma sap |
Loai | Dien tích | Tien / m2 | Tien mat bang | Phi khac | Tong cong -----
--- 1 | A724 | Thuc pham | 78 m2 | 560.000 | 43.680.000 | 3.000.000 |
46.680.000 d 2 | B419 | Quan ao | 33 m2 | 560.000 | 18.480.000 | 0 |
18.480.000 d 3 | C682 | Da qui | 55 m2 | 560.000 | 30.800.000 | 7.000.000 |
37.800.000 d 4 | B331 | Quan ao | 38 m2 | 560.000 | 21.280.000 | 0 |
21.280.000 d 5 | A903 | Thuc pham | 60 m2 | 560.000 | 33.600.000 | 5.000.000
| 38.600.000 d Tong tien thue mat bang thu trong thang: 167.000.000 d
```

Submission instructions

Compress all your work into a file using the following format: **StudentID.zip/rar** and submit this file back into Moodle.